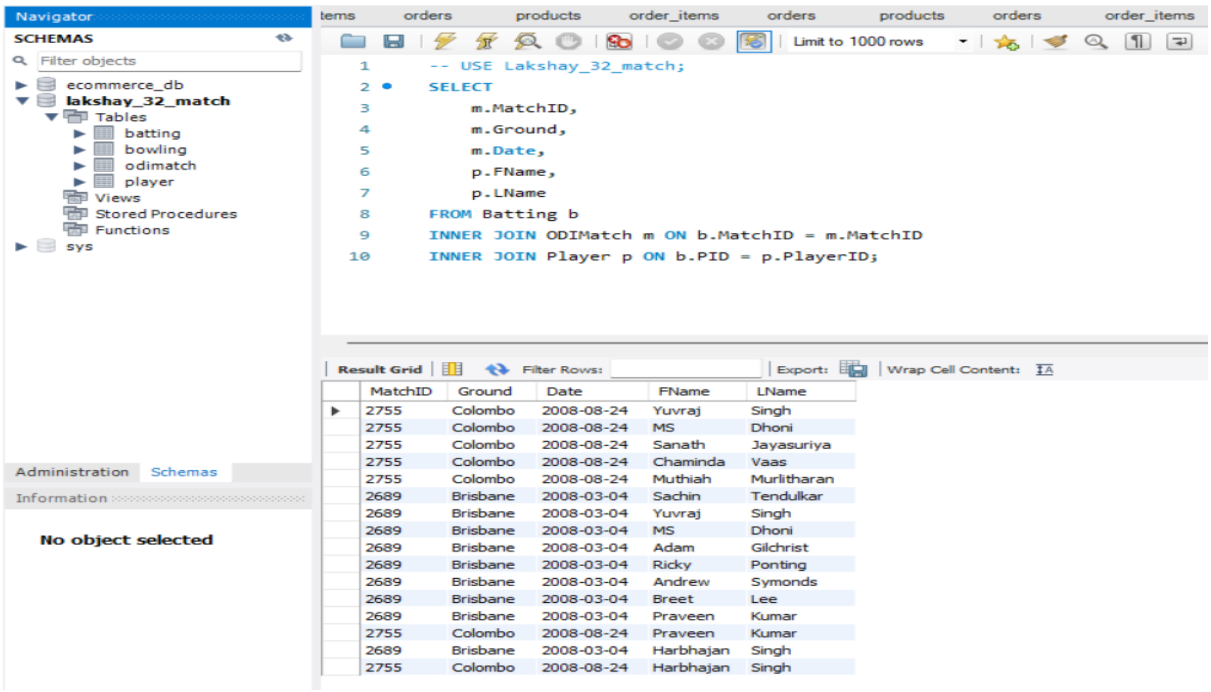
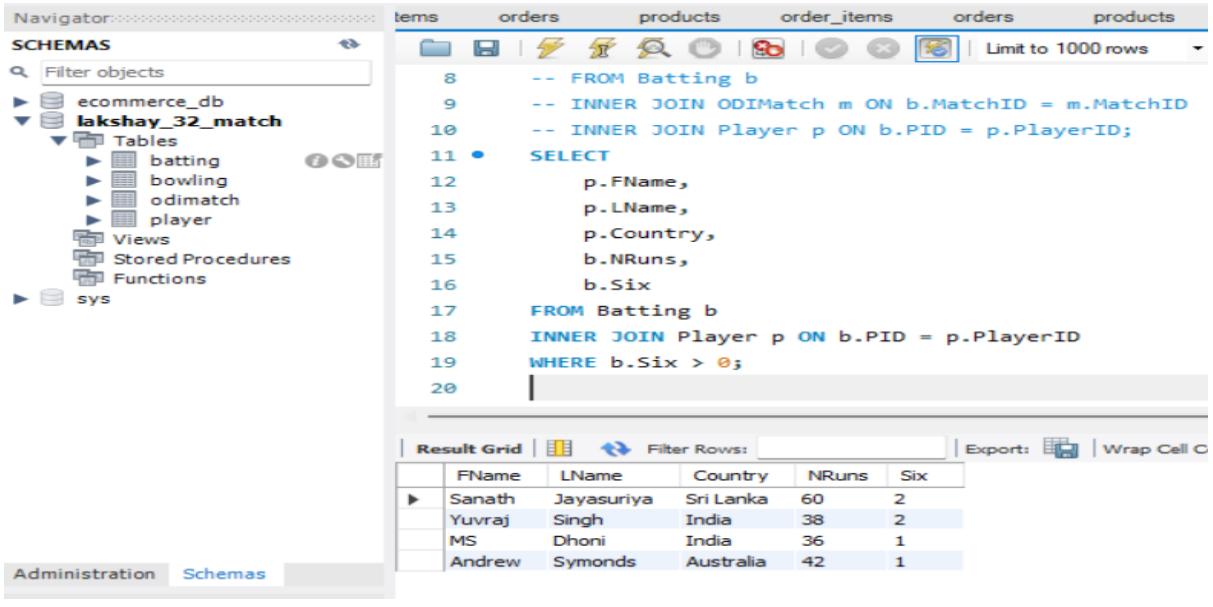
LAB 6: Implementation of Various Join Operations

Aim: Understanding and applying INNER JOIN, LEFT OUTER JOIN, RIGHT OUTER JOIN, FULL OUTER JOIN, SELF JOIN, CROSS JOIN, and Multi-table Joins on relational schemas.

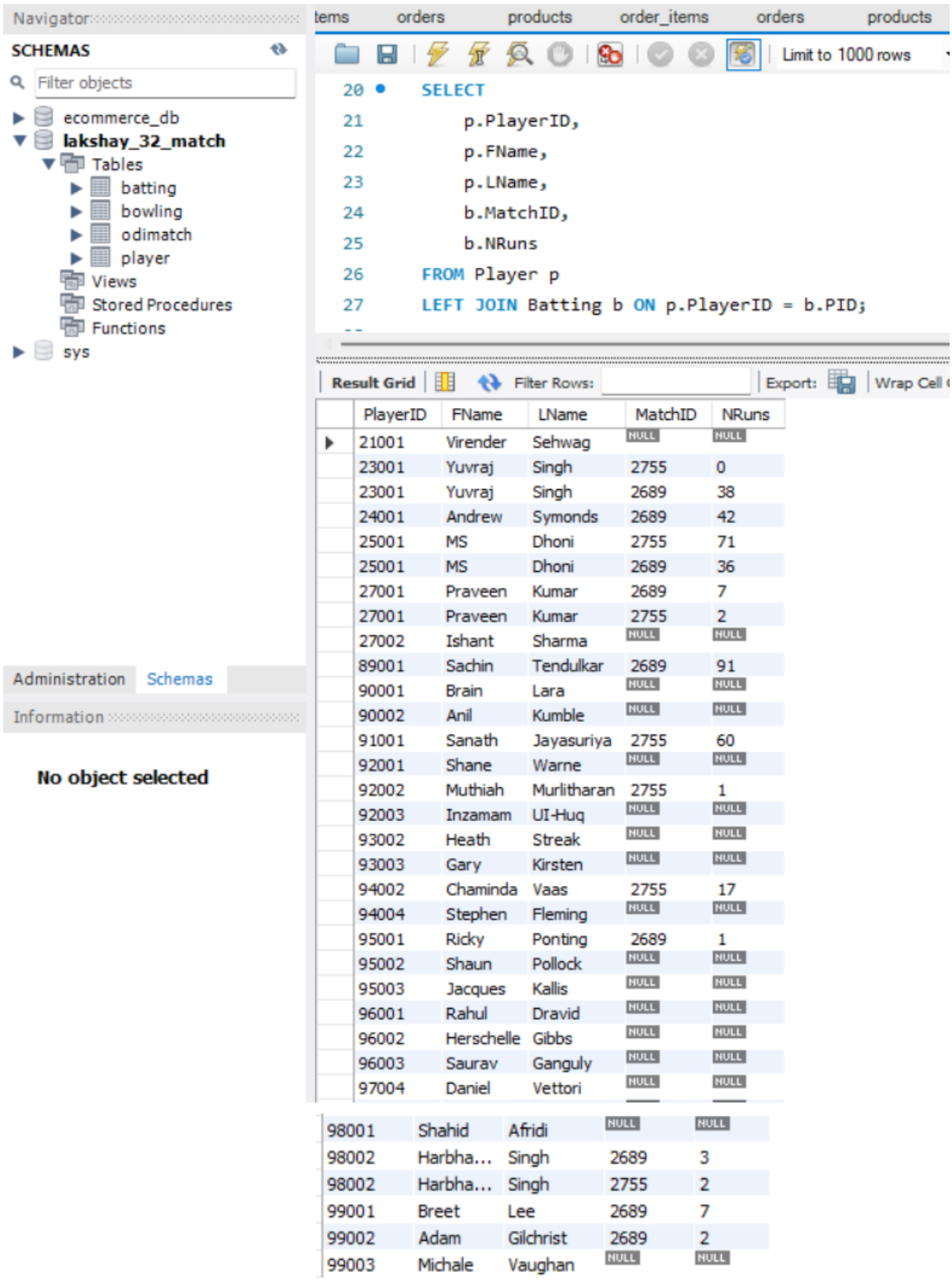
1 [Inner Join]: Write an SQL query using INNER JOIN to display the MatchID, Ground, Date, and the Player's full name (FName, LName) for every player who batted in each match.



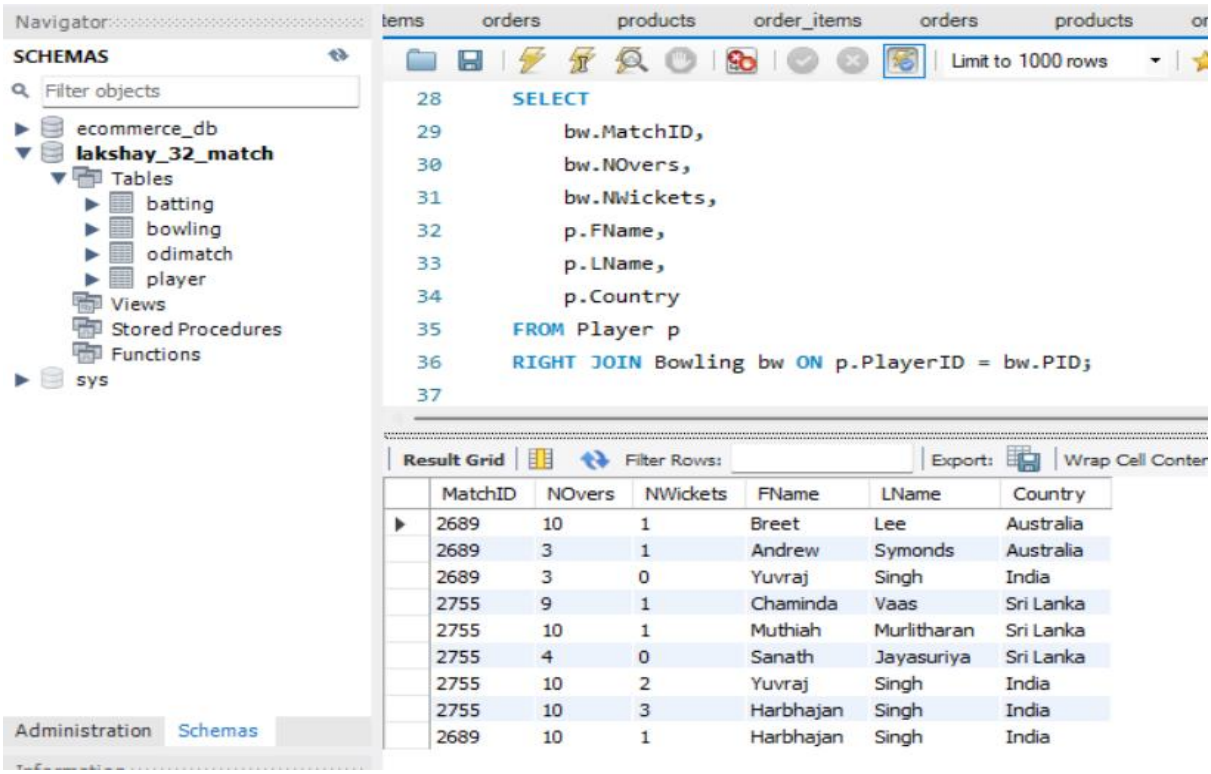
2 [Inner Join with Filter]: Display Player full name, Country, NRuns, and Six for all batting performances where the player scored at least one six (Six > 0).



3 [Left Outer Join]: Write a query using LEFT OUTER JOIN to list all players (PlayerID, FName, LName) along with their batting scores (MatchID, NRuns). Ensure players who have never batted in any match are also displayed with NULL values.

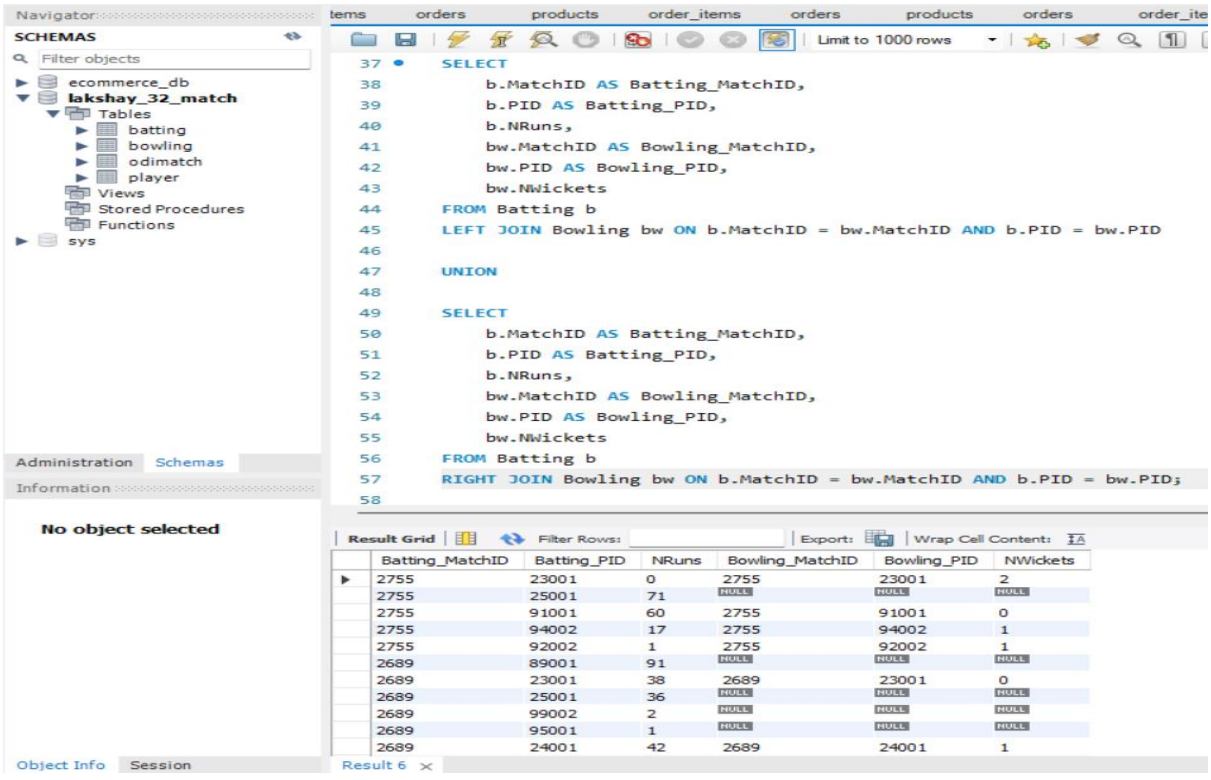


4 [Right Outer Join]: Retrieve all bowling performance entries (MatchID, NOvers, Nwickets) along with player details (FName, LName, Country) using a RIGHT OUTER JOIN, showing all bowling entries even if player profiles are missing.



MatchID	NOvers	NWickets	FName	LName	Country
2689	10	1	Breet	Lee	Australia
2689	3	1	Andrew	Symonds	Australia
2689	3	0	Yuvraj	Singh	India
2755	9	1	Chaminda	Vaas	Sri Lanka
2755	10	1	Muthiah	Murlitharan	Sri Lanka
2755	4	0	Sanath	Jayasuriya	Sri Lanka
2755	10	2	Yuvraj	Singh	India
2755	10	3	Harbhajan	Singh	India
2689	10	1	Harbhajan	Singh	India

5 [Full Outer Join / Union]: Write an SQL query to perform a FULL OUTER JOIN (or simulate using LEFT JOIN UNION RIGHT JOIN) between Batting and Bowling tables on MatchID and PID to view all combined match participation records.

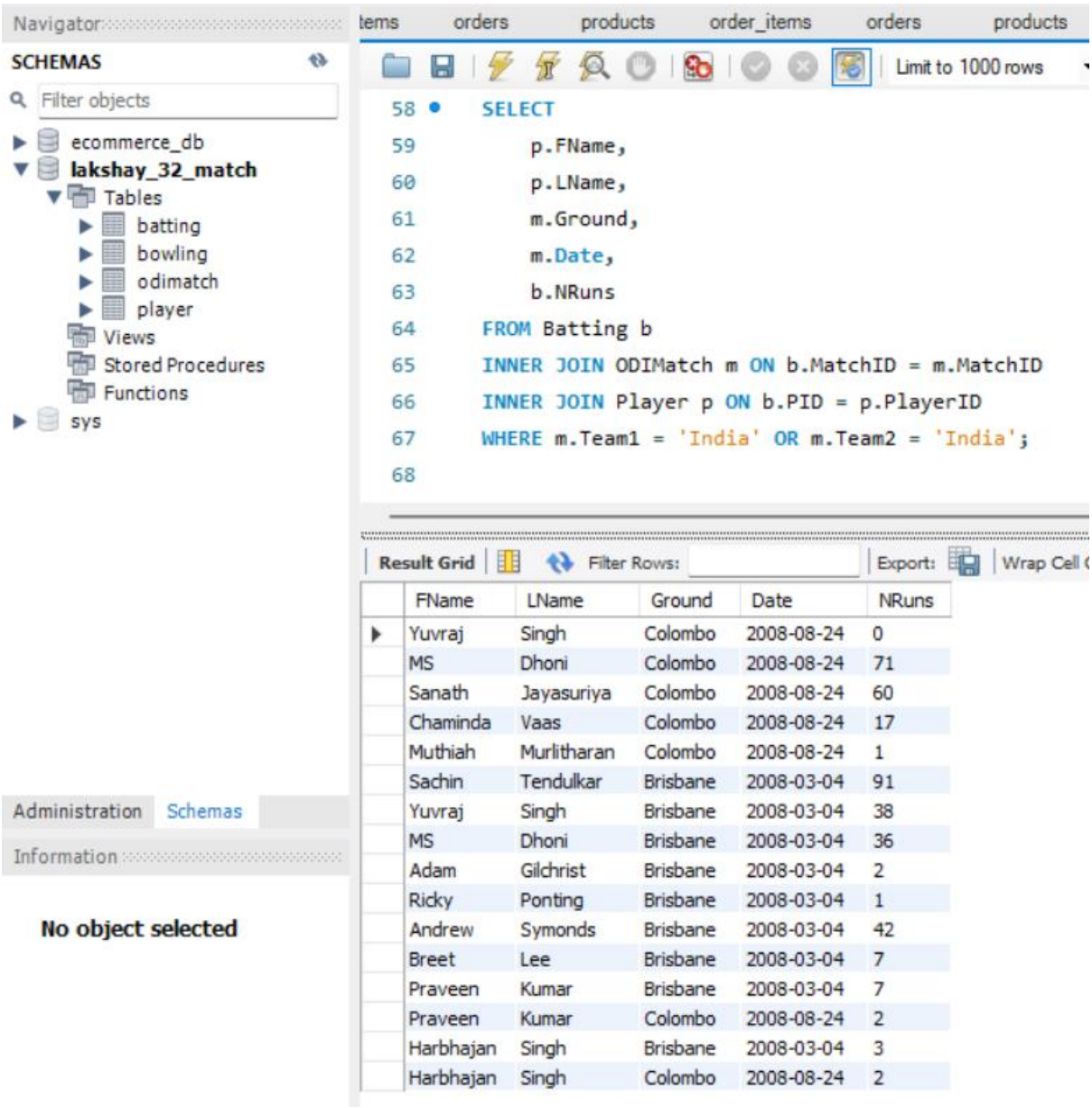


Batting_MatchID	Batting_PID	NRuns	Bowling_MatchID	Bowling_PID	NWickets
2755	23001	0	2755	23001	2
2755	25001	71	NULL	NULL	NULL
2755	91001	60	2755	91001	0
2755	94002	17	2755	94002	1
2755	92002	1	2755	92002	1
2689	89001	91	NULL	NULL	NULL
2689	23001	38	2689	23001	0
2689	25001	36	NULL	NULL	NULL
2689	99002	2	NULL	NULL	NULL
2689	95001	1	NULL	NULL	NULL
2689	24001	42	2689	24001	1

2689	99001	7	2689	99001	1
2689	27001	7	NULL	NULL	NULL
2755	27001	2	NULL	NULL	NULL
2689	98002	3	2689	98002	1
2755	98002	2	2755	98002	3

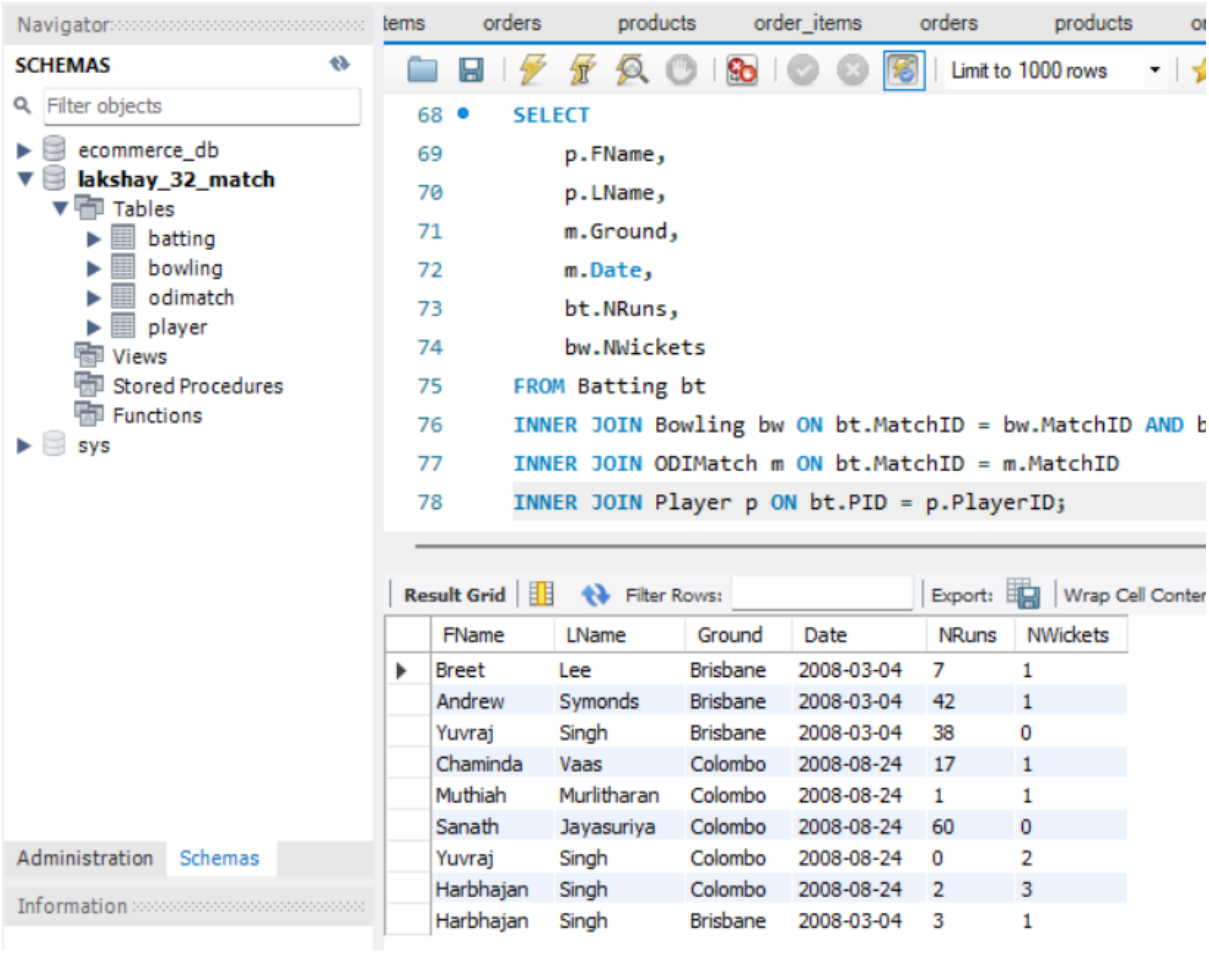
Object Info | Session | Result 6

6 [3-Table Inner Join]: List Player.FName, Player.LName, ODIMatch.Ground, ODIMatch.Date, and Batting.NRuns for all matches played where 'India' was either Team1 or Team2.

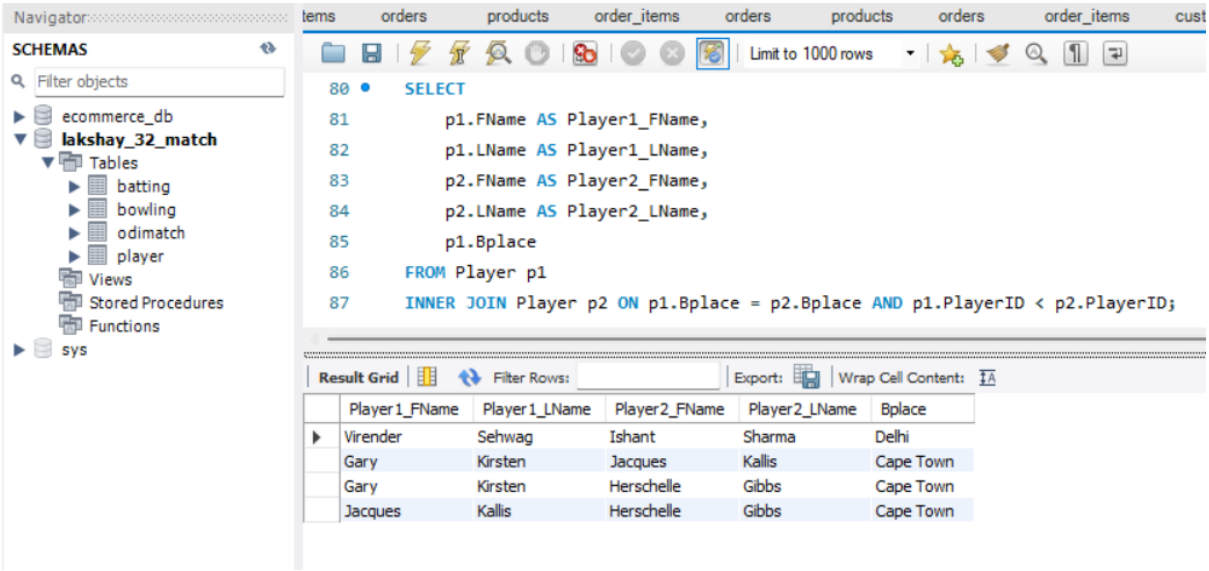


FName	LName	Ground	Date	NRuns
Yuvraj	Singh	Colombo	2008-08-24	0
MS	Dhoni	Colombo	2008-08-24	71
Sanath	Jayasuriya	Colombo	2008-08-24	60
Chaminda	Vaas	Colombo	2008-08-24	17
Muthiah	Murlitharan	Colombo	2008-08-24	1
Sachin	Tendulkar	Brisbane	2008-03-04	91
Yuvraj	Singh	Brisbane	2008-03-04	38
MS	Dhoni	Brisbane	2008-03-04	36
Adam	Gilchrist	Brisbane	2008-03-04	2
Ricky	Ponting	Brisbane	2008-03-04	1
Andrew	Symonds	Brisbane	2008-03-04	42
Breet	Lee	Brisbane	2008-03-04	7
Praveen	Kumar	Brisbane	2008-03-04	7
Praveen	Kumar	Colombo	2008-08-24	2
Harbhajan	Singh	Brisbane	2008-03-04	3
Harbhajan	Singh	Colombo	2008-08-24	2

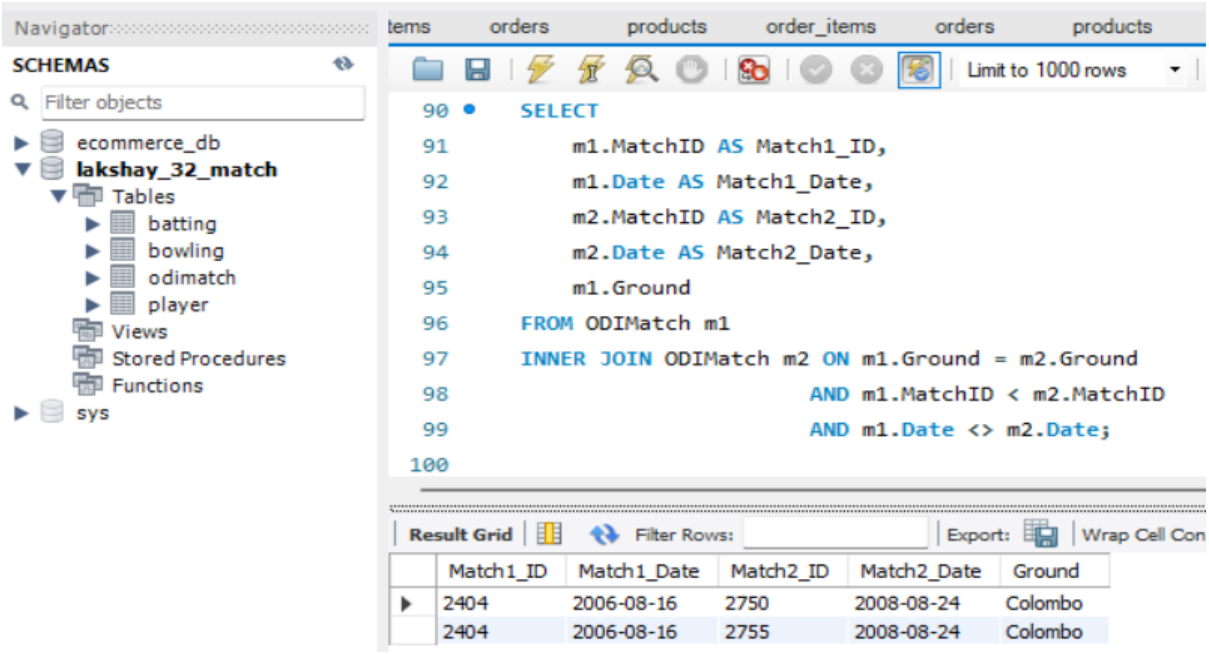
7 [4-Table Multi-Join]: Display player full name, Ground, Date, Batting.NRuns, and Bowling.Nwickets for all players who both BATTED and BOWLED in the exact same match.



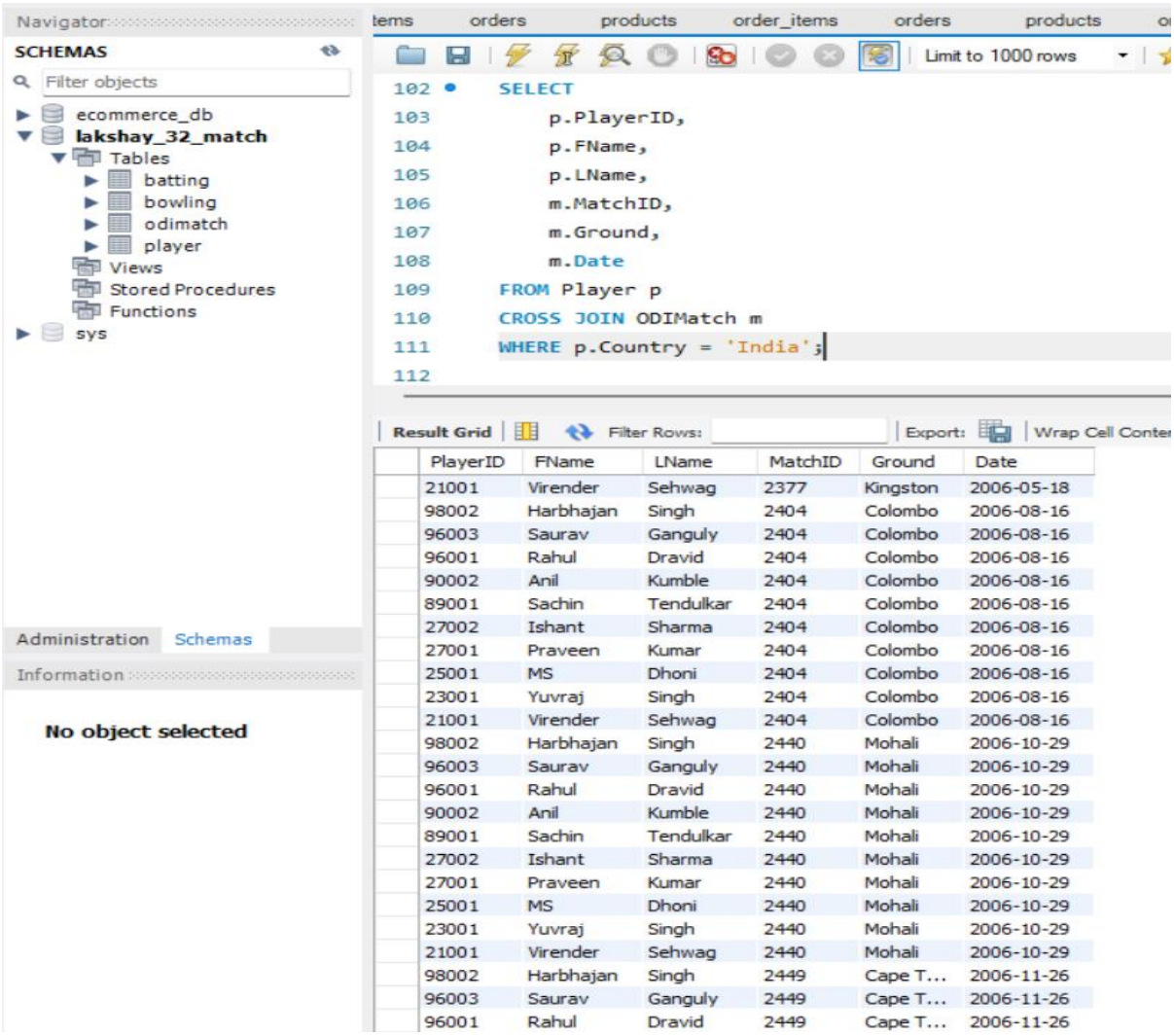
8 [Self Join - Attribute Match]: Write a query using a SELF JOIN on the Player table to find pairs of players who were born in the same birth place (Bplace). Display Player 1 Name, Player 2 Name, and Bplace (exclude self-pairings and duplicate reversed pairs).

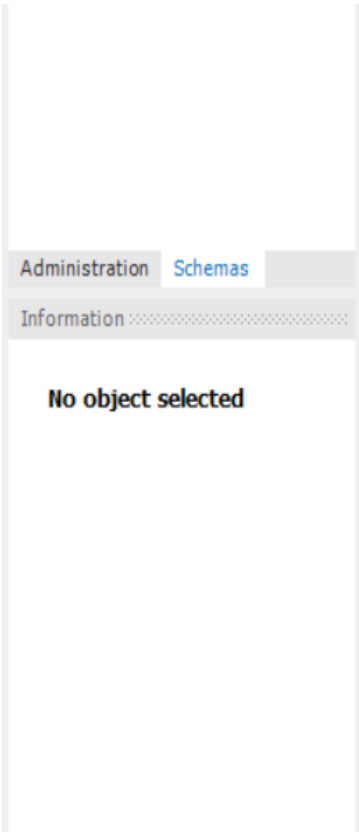


9 [Self Join - Match Analysis]: Using a SELF JOIN on ODIMatch, find pairs of distinct matches that were played on the same Ground on different dates.

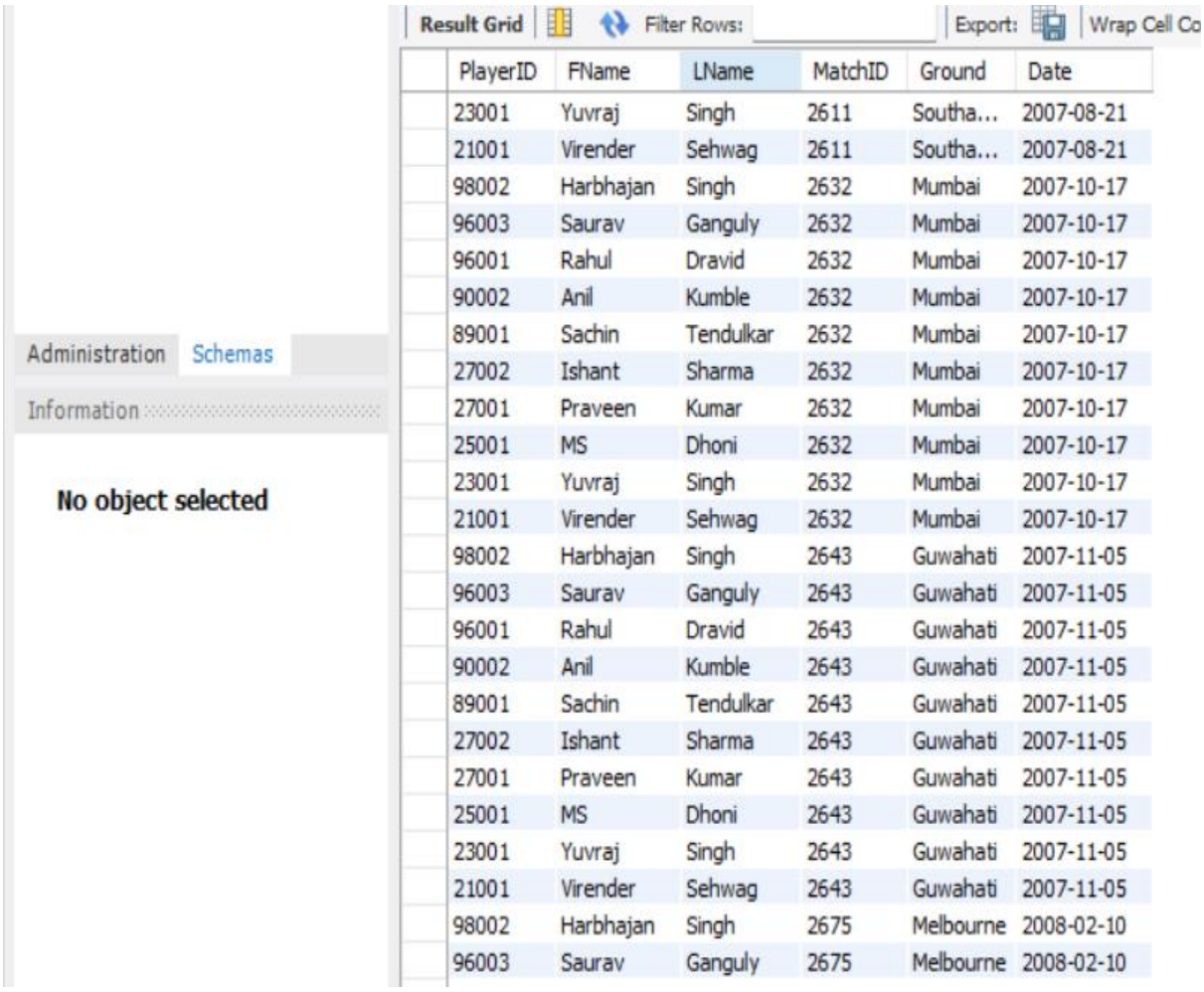


10 [Cross Join]: Generate a Cartesian product (CROSS JOIN) between players from 'India' and all matches in ODIMatch to display all hypothetical player-match combinations.





PlayerID	FName	LName	MatchID	Ground	Date
89001	Sachin	Tendulkar	2493	Vadodara	2007-01-31
27002	Ishant	Sharma	2493	Vadodara	2007-01-31
27001	Praveen	Kumar	2493	Vadodara	2007-01-31
25001	MS	Dhoni	2493	Vadodara	2007-01-31
23001	Yuvraj	Singh	2493	Vadodara	2007-01-31
21001	Virender	Sehwag	2493	Vadodara	2007-01-31
98002	Harbhajan	Singh	2520	Rajkot	2007-02-11
96003	Saurav	Ganguly	2520	Rajkot	2007-02-11
96001	Rahul	Dravid	2520	Rajkot	2007-02-11
90002	Anil	Kumble	2520	Rajkot	2007-02-11
89001	Sachin	Tendulkar	2520	Rajkot	2007-02-11
27002	Ishant	Sharma	2520	Rajkot	2007-02-11
27001	Praveen	Kumar	2520	Rajkot	2007-02-11
25001	MS	Dhoni	2520	Rajkot	2007-02-11
23001	Yuvraj	Singh	2520	Rajkot	2007-02-11
21001	Virender	Sehwag	2520	Rajkot	2007-02-11
98002	Harbhajan	Singh	2611	Southa...	2007-08-21
96003	Saurav	Ganguly	2611	Southa...	2007-08-21
96001	Rahul	Dravid	2611	Southa...	2007-08-21
90002	Anil	Kumble	2611	Southa...	2007-08-21
89001	Sachin	Tendulkar	2611	Southa...	2007-08-21
27002	Ishant	Sharma	2611	Southa...	2007-08-21
27001	Praveen	Kumar	2611	Southa...	2007-08-21
25001	MS	Dhoni	2611	Southa...	2007-08-21



PlayerID	FName	LName	MatchID	Ground	Date
23001	Yuvraj	Singh	2611	Southa...	2007-08-21
21001	Virender	Sehwag	2611	Southa...	2007-08-21
98002	Harbhajan	Singh	2632	Mumbai	2007-10-17
96003	Saurav	Ganguly	2632	Mumbai	2007-10-17
96001	Rahul	Dravid	2632	Mumbai	2007-10-17
90002	Anil	Kumble	2632	Mumbai	2007-10-17
89001	Sachin	Tendulkar	2632	Mumbai	2007-10-17
27002	Ishant	Sharma	2632	Mumbai	2007-10-17
27001	Praveen	Kumar	2632	Mumbai	2007-10-17
25001	MS	Dhoni	2632	Mumbai	2007-10-17
23001	Yuvraj	Singh	2632	Mumbai	2007-10-17
21001	Virender	Sehwag	2632	Mumbai	2007-10-17
98002	Harbhajan	Singh	2643	Guwahati	2007-11-05
96003	Saurav	Ganguly	2643	Guwahati	2007-11-05
96001	Rahul	Dravid	2643	Guwahati	2007-11-05
90002	Anil	Kumble	2643	Guwahati	2007-11-05
89001	Sachin	Tendulkar	2643	Guwahati	2007-11-05
27002	Ishant	Sharma	2643	Guwahati	2007-11-05
27001	Praveen	Kumar	2643	Guwahati	2007-11-05
25001	MS	Dhoni	2643	Guwahati	2007-11-05
23001	Yuvraj	Singh	2643	Guwahati	2007-11-05
21001	Virender	Sehwag	2643	Guwahati	2007-11-05
98002	Harbhajan	Singh	2675	Melbourne	2008-02-10
96003	Saurav	Ganguly	2675	Melbourne	2008-02-10

Administration Schemas

Information

No object selected

PlayerID	FName	LName	MatchID	Ground	Date
96001	Rahul	Dravid	2675	Melbourne	2008-02-10
90002	Anil	Kumble	2675	Melbourne	2008-02-10
89001	Sachin	Tendulkar	2675	Melbourne	2008-02-10
27002	Ishant	Sharma	2675	Melbourne	2008-02-10
27001	Praveen	Kumar	2675	Melbourne	2008-02-10
25001	MS	Dhoni	2675	Melbourne	2008-02-10
23001	Yuvraj	Singh	2675	Melbourne	2008-02-10
21001	Virender	Sehwag	2675	Melbourne	2008-02-10
98002	Harbhajan	Singh	2681	Adelaide	2008-02-19
96003	Saurav	Ganguly	2681	Adelaide	2008-02-19
96001	Rahul	Dravid	2681	Adelaide	2008-02-19
90002	Anil	Kumble	2681	Adelaide	2008-02-19
89001	Sachin	Tendulkar	2681	Adelaide	2008-02-19
27002	Ishant	Sharma	2681	Adelaide	2008-02-19
27001	Praveen	Kumar	2681	Adelaide	2008-02-19
25001	MS	Dhoni	2681	Adelaide	2008-02-19
23001	Yuvraj	Singh	2681	Adelaide	2008-02-19
21001	Virender	Sehwag	2681	Adelaide	2008-02-19
98002	Harbhajan	Singh	2688	Sydney	2008-03-02
96003	Saurav	Ganguly	2688	Sydney	2008-03-02
96001	Rahul	Dravid	2688	Sydney	2008-03-02
90002	Anil	Kumble	2688	Sydney	2008-03-02
89001	Sachin	Tendulkar	2688	Sydney	2008-03-02
27002	Ishant	Sharma	2688	Sydney	2008-03-02

Administration Schemas

Information

No object selected

PlayerID	FName	LName	MatchID	Ground	Date
27001	Praveen	Kumar	2688	Sydney	2008-03-02
25001	MS	Dhoni	2688	Sydney	2008-03-02
23001	Yuvraj	Singh	2688	Sydney	2008-03-02
21001	Virender	Sehwag	2688	Sydney	2008-03-02
98002	Harbhajan	Singh	2689	Brisbane	2008-03-04
96003	Saurav	Ganguly	2689	Brisbane	2008-03-04
96001	Rahul	Dravid	2689	Brisbane	2008-03-04
90002	Anil	Kumble	2689	Brisbane	2008-03-04
89001	Sachin	Tendulkar	2689	Brisbane	2008-03-04
27002	Ishant	Sharma	2689	Brisbane	2008-03-04
27001	Praveen	Kumar	2689	Brisbane	2008-03-04
25001	MS	Dhoni	2689	Brisbane	2008-03-04
23001	Yuvraj	Singh	2689	Brisbane	2008-03-04
21001	Virender	Sehwag	2689	Brisbane	2008-03-04
98002	Harbhajan	Singh	2717	Karachi	2008-06-26
96003	Saurav	Ganguly	2717	Karachi	2008-06-26
96001	Rahul	Dravid	2717	Karachi	2008-06-26
90002	Anil	Kumble	2717	Karachi	2008-06-26
89001	Sachin	Tendulkar	2717	Karachi	2008-06-26
27002	Ishant	Sharma	2717	Karachi	2008-06-26
27001	Praveen	Kumar	2717	Karachi	2008-06-26
25001	MS	Dhoni	2717	Karachi	2008-06-26
23001	Yuvraj	Singh	2717	Karachi	2008-06-26
21001	Virender	Sehwag	2717	Karachi	2008-06-26

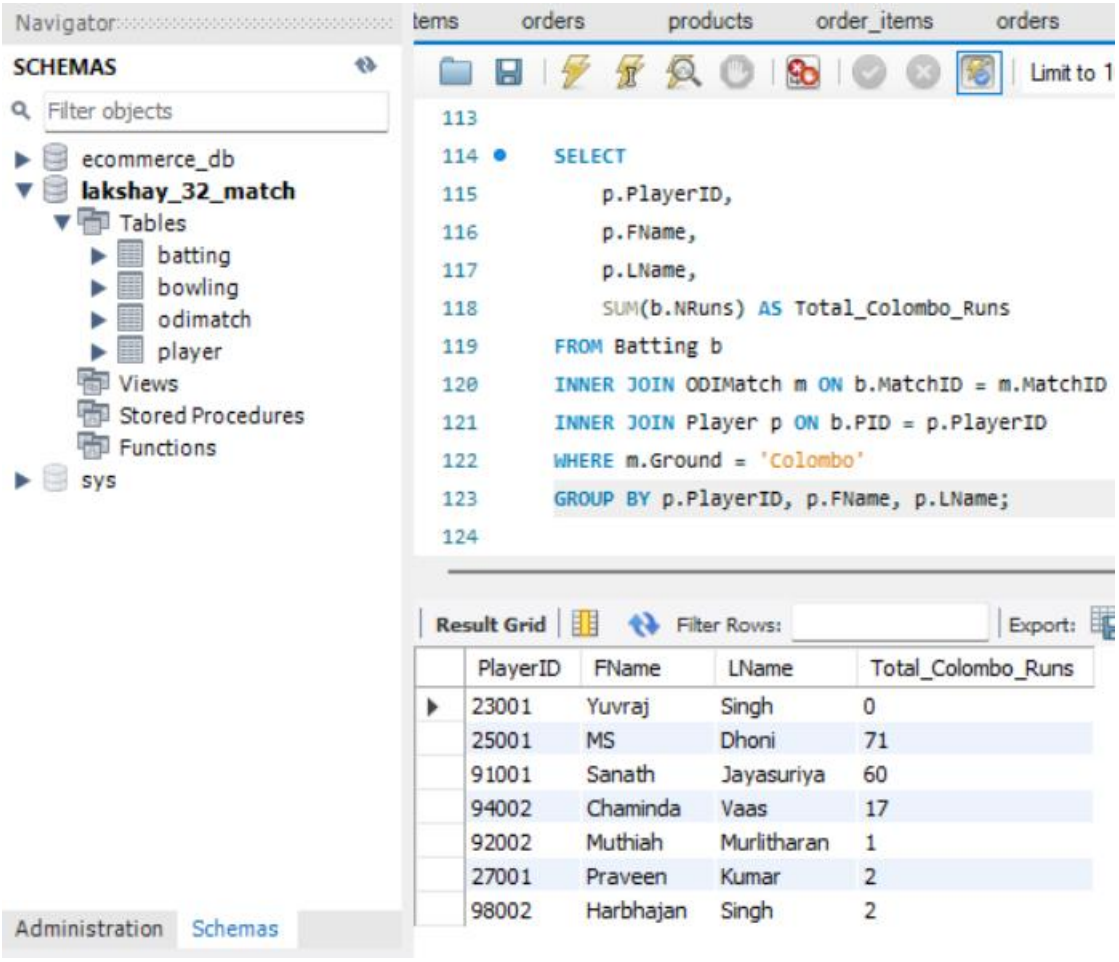
Administration Schemas

Information

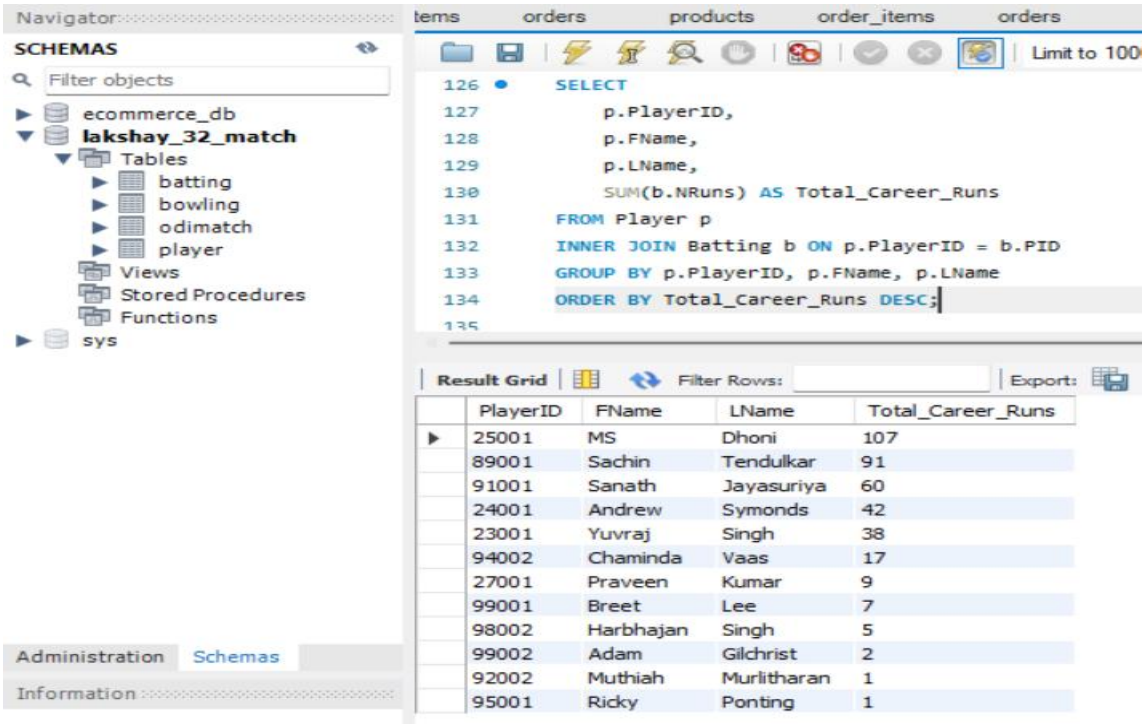
No object selected

21001	Virender	Sehwag	2717	Karachi	2008-06-26
98002	Harbhajan	Singh	2750	Colombo	2008-08-24
96003	Saurav	Ganguly	2750	Colombo	2008-08-24
96001	Rahul	Dravid	2750	Colombo	2008-08-24
90002	Anil	Kumble	2750	Colombo	2008-08-24
89001	Sachin	Tendulkar	2750	Colombo	2008-08-24
27002	Ishant	Sharma	2750	Colombo	2008-08-24
27001	Praveen	Kumar	2750	Colombo	2008-08-24
25001	MS	Dhoni	2750	Colombo	2008-08-24
23001	Yuvraj	Singh	2750	Colombo	2008-08-24
21001	Virender	Sehwag	2750	Colombo	2008-08-24
98002	Harbhajan	Singh	2755	Colombo	2008-08-24
96003	Saurav	Ganguly	2755	Colombo	2008-08-24
96001	Rahul	Dravid	2755	Colombo	2008-08-24
90002	Anil	Kumble	2755	Colombo	2008-08-24
89001	Sachin	Tendulkar	2755	Colombo	2008-08-24
27002	Ishant	Sharma	2755	Colombo	2008-08-24
27001	Praveen	Kumar	2755	Colombo	2008-08-24
25001	MS	Dhoni	2755	Colombo	2008-08-24
23001	Yuvraj	Singh	2755	Colombo	2008-08-24
21001	Virender	Sehwag	2755	Colombo	2008-08-24

11 [Natural Join vs Explicit]: Write an SQL query joining Batting and ODIMatch using MatchID to display total runs scored by each player in matches played at 'Colombo' ground.



12 [Join with Aggregation]: Display PlayerID, FName, LName, and total career runs (SUM(NRuns)) across all matches by joining Player and Batting, sorted in descending order of total runs.



13 [Left Join Anti-Join]: Write an SQL query using LEFT JOIN to identify all players (PlayerID, FName, LName, Country) who have NEVER recorded a bowling performance in the Bowling table.

